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Unveiling Molecular Interactions in Chocolate by Molecular Dynamics: A Study on Caffeine, Fatty Acids, and Sugars

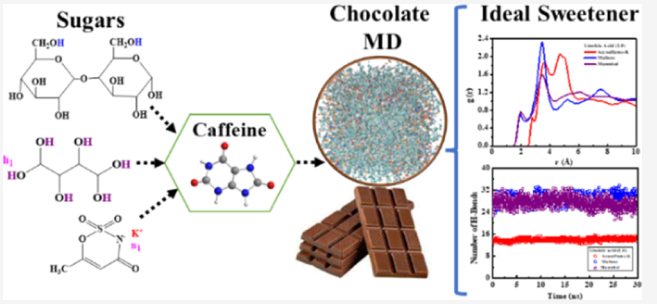
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ABSTRACT: This study investigates the incorporation of caffeine into chocolate protein bars, focusing on its interactions with fatty acids from cocoa powder and various sweeteners, including Acesulfame-K, Maltose, and Mannitol. Utilizing molecular dynamics simulations, hydrogen bonding, radial distribution function, and mean squared displacement are analyzed to understand the molecular behavior and interactions at the atomic level. The study reveals that caffeine's interaction with fatty acids significantly influences the stability and efficacy of the final product. The choice of sweeteners also impacts these interactions, with each sugar demonstrating unique properties that affect the dynamics within the chocolate matrix. The simulations show subdiffusive to diffusive transitions, indicating strong initial binding that gradually loosens over time. This detailed molecular insight is crucial for optimizing the formulation, processing conditions, and storage of caffeine-fortified chocolate protein bars, ultimately enhancing product quality and bridging the knowledge gap in understanding how these molecular interactions affect the properties of chocolate protein bars.



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1. INTRODUCTION

The incorporation of caffeine into functional foods such as chocolate protein bars has gained significant attention due to its potential health benefits and its role as a stimulant. Caffeine, a naturally occurring alkaloid, is widely consumed globally and is known for its effects on the central nervous system, leading to enhanced alertness and cognitive performance.[1] The use of extracted caffeine in chocolate protein bars not only enhances the nutritional profile but also offers added value to consumers seeking both protein and a stimulant in a single product.[2] The interaction of caffeine with the various constituents of cocoa powder, particularly fatty acids, is crucial to determine the overall stability and efficacy of the final product. Cocoa powder is rich in both saturated and unsaturated fatty acids, which play a significant role in the texture, flavor, and stability of chocolate products.[3] Understanding the interaction between caffeine, fatty acids, and sweeteners is essential for optimizing the formulation of chocolate protein bars.

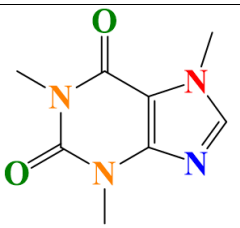
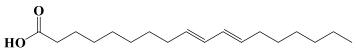
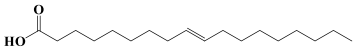
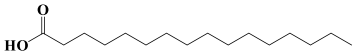
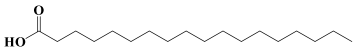
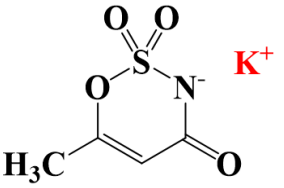
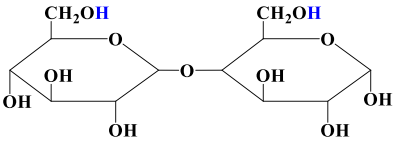
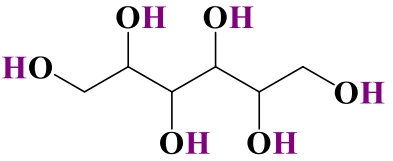
Moreover, the choice of sweeteners can influence the interaction of caffeine with cocoa constituents. The interaction of sweeteners with caffeine and cocoa constituents can alter the physicochemical and sensory properties of cocoa-based beverages. Sweeteners influence the solubility, dispersion, and cohesion of cocoa mixtures, affecting the release and perception of caffeine. Alternative sweeteners, such as stevia or aspartame, may modify the matrix of cocoa mixtures due to their distinct molecular structures, potentially enhancing or diminishing interactions with caffeine. Sweeteners that increase the solubility of cocoa powders can promote better caffeine dispersion, while others might introduce binding effects with polyphenolic compounds, affecting bioavailability.[4,5] Additionally, sweeteners with herbal or bitter aftertastes, such as stevia, may synergize or mask the natural bitterness of caffeine, influencing sensory attributes. Sweetener choice also affects the antioxidant content and activity in cocoa drinks. Mixtures with lower fat content and specific sweeteners often exhibit higher polyphenol levels, which can interact with caffeine, modifying its absorption and metabolism. Sweeteners like inulin or fructose enhance sweetness perception while maintaining an acceptable mouthfeel, balancing the bitterness of caffeine and enriching the overall flavor profile.[6] The observed sensory and chemical modifications suggest that selecting appropriate sweeteners can optimize the interplay of caffeine with cocoa

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Table 1. System Composition

Compound Name	Abbreviation Abbreviation	Structure	CAS NO.	References
Caffeine	CA0		58-08-2	25
Linoleic Acid	L0		60-33-3	26
Oleic Acid	OL0		112-80-1	27
Palmitic acid	P0		57-10-3	28
Stearic Acid	S0		57-11-4	29
Acesulfame-K			55589-62-3	30
Maltose			69-79-4	31
Mannitol			69-65-8	32

constituents, improving both health-related benefits and consumer acceptability of cocoa-based beverages.[5,7] Experimental studies have shown that different sugars can significantly affect the texture, stability, and sensory properties of chocolate.[8−11] Despite the advances in understanding the individual components of chocolate and caffeine interactions, there remains a knowledge gap in comprehensively understanding how these interactions affect the final product's stability. This study aims to fill this gap by providing detailed molecular insights that are difficult to obtain experimentally. The implications of these findings extend to the practical aspects of chocolate protein bar production. Understanding the molecular interactions can guide the selection of ingredients to enhance the product's stability and consumer satisfaction. This study focuses on three different sugars: Acesulfame-K, Maltose, and Mannitol. Each of these sugars has unique properties that can affect the molecular interactions within the chocolate protein bar matrix.[12]

Molecular dynamics (MD) simulations provide detailed insights into the behavior of molecules at the atomic level, allowing for a comprehensive understanding of the molecular interactions in complex systems.[13] Furthermore, the application of MD simulations in this context exemplifies the potential of computational techniques in food science, offering a powerful tool to predict and optimize product formulations without extensive trial-and-error in experimental setups.[13] MD simulations provide detailed insights into the structural and dynamic behaviors of chocolate during processing, particularly

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in conching. By modeling chocolate's primary components—cocoa butter, sucrose, and lecithin—MD simulations reveal molecular interactions that are critical to its rheological and textural properties. Cocoa butter is represented by triglycerides such as POS, POP, and SOS, while emulsifiers such as DLPC highlight hydrogen bonding with sucrose. Simulations demonstrate the formation of phospholipid monolayers at sucrose interfaces, impacting viscosity and yield stress.[14] Micelle formation in cocoa butter and temperature-driven phase transitions are explored, linking molecular structures to macroscopic phenomena, such as fat bloom. Techniques such as pull-code simulations elucidate the desorption mechanics of emulsifiers from sugar crystals, showing the role of hydrogen bonds and viscous resistance. The Avrami model describes nucleation and growth of surface patterns, correlating simulation and experimental observations.[15] Despite challenges like computational limits and slow diffusion in viscous media, MD simulations effectively capture atomistic details and dynamic behaviors.[16] These findings enhance the understanding of chocolate's micro- and nanostructures, aiding in optimizing formulations and processes. MD simulations thus emerge as a powerful tool in food research, bridging experimental gaps and advancing the molecular-level comprehension of complex food systems.

The present study employs MD simulations to investigate the interactions among caffeine, fatty acids from cocoa powder, and the selected sugars. The analysis of these interactions is performed using various computational techniques, such as hydrogen bonding (H-bonding) analysis, radial distribution function (RDF), and mean squared displacement (MSD). H-bonding is a critical factor in determining the stability and interaction strength between molecules. RDF analysis provides information on the spatial distribution of molecules, which is essential for understanding how caffeine and other constituents are organized within the chocolate matrix.[17] MSD analysis helps in assessing the mobility of molecules, which can influence the stability of the product.[18] Experimentally investigating these interactions poses several challenges. One significant challenge is the complexity of isolating and characterizing the individual interactions among caffeine, fatty acids, and sugars in a heterogeneous matrix like a chocolate protein bar. Additionally, the dynamic nature of these interactions and the influence of various processing conditions, such as temperature and mixing, further complicate experimental analysis.[18] These challenges highlight the benefits of using MD simulations, which can model and predict molecular behavior under various conditions without the need for expensive and time-consuming experimental setup.

2. COMPUTATIONAL METHODOLOGY

Primarily, chocolate protein bars contain sugars, cocoa butter, fatty acids (saturated and unsaturated), protein, etc. The respective structures are given in Table 1 and Figure S1, given the labeled caffeine structure.[25−32] Sugars such as Acesulfame-K, Maltose, and Mannitol are selected for the study. The structures are initially drawn in the Avogadro package,[19] followed by the usage of the PACKMOL package,[20] to create the simulation boxes for the MD simulation. The Optimized Potentials for Liquid Simulations (OPLS) force field has been proven to be an effective force field for organic molecule extraction using polar and neoteric solvents[21] and henceforth is used in this work. The functional form of the OPLS force field contains bonded (bond, angle, dihedral, and improper) and nonbonded (6−12 LJ potential and Coulombic) energies, whose functional form is described in the literature.[22−24] The details of the systems are given in Table S1 of the Supporting Information.

2.1. Methodology for MD Simulation. The MD simulation comprises various steps such as energy minimization, NVT and NPT ensembles, which contribute to the equilibration run, and finally, a production run. In this study, every MD simulation is performed in the GROMACS package,[33] whereas the Visual Molecular Dynamics (VMD) package employs the trajectory analysis.[34] The use of energy minimization is to lower the potential energy of the system, which helps molecules attain a stable conformation and also enhances the stability of the system. The steepest descent algorithm with a convergence threshold of 1000 kJ/mol•nm is used for energy minimization. It is presently underway for 500,000 steps, having a step size equal to 0.001 nm. A leapfrog integrator[35] integrates all equations of motion using a time step of 2 fs during the equilibration and final production run. The linear constraint solver (LINCS) algorithm[36] constrains hydrogen bonds in the equilibration stage. The particle-mesh Ewald method[37] handles the long-range electrostatic interactions. The van der Waals (vdW) interactions are limited to a distance of 1.2 nm. The Verlet algorithm updates the neighbor grid list at a frequency of 20 steps. Equilibration occurs in two stages: the first stage involves a canonical ensemble (NVT), and the second stage involves an isothermal isobaric ensemble (NPT). Throughout both the equilibration and final production stages, the temperature remains constant via the velocity-rescaling thermostat (V-rescale).[20] Additionally, the pressure remains controlled during the NPT and final production stages by employing the Parrinello−Rahman barostat.[38] A unit nanosecond simulation with the NVT ensemble is conducted at a temperature of 310 K with a time step of 2 fs. The simulation with the NPT ensemble runs for 19 ns at a temperature of 310 K and a pressure of 1.013 bar, with a time step of 2 fs. The outcome of NPT equilibration indicates that the energy, density, and volume of the system remain nearly constant. This finding suggests that the system is adequately equilibrated and ready for production run. The final production run spans 30 ns at temperatures of 310 K and pressures of 1.013 bar. Coordinates and velocities of the system are recorded every 10 ps for future analysis.

2.2. Analysis of Trajectories. *2.2.1. Hydrogen Bonding (H-Bonding).* The analysis focuses on H-bonding interactions among solutes and sugars, which are pivotal in extraction behavior. The strength of H-bonds among caffeine, other compounds, and solvents is influenced by the sugar type, interaction extent, and the functional groups capable of forming H-bonds. A H-bond forms when the hydrogen donor-to-acceptor angle (α_{HDA}) is less than 180°−150° and the donor−acceptor distance (r_{DA}) is less than 0.35 nm.[39] The intermittent H-bond autocorrelation function is calculated to understand the temporal evolution of H-bonds within the system.

$$C_{HB} = \frac{\langle H(\tau)H(\tau + t)\rangle}{\langle H(\tau)H(\tau)\rangle} \quad (1)$$

In this context, $H(t)$ is considered to be either 1 or 0. C_{HB} represents the count of hydrogen bonds and depends on whether the selected pair forms a hydrogen bond at that particular moment in time.

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2.2.2. Radial Distribution Function. The RDF is used to understand the spatial distribution of atoms around a reference atom, providing insights into structural information and atom−atom interactions. The RDF, denoted as $g(r)$, is mathematically expressed as

$$g(r) = \frac{1}{N}\left(\frac{V}{4\pi r^2 dr}\right)\sum_{i=1}^{N}\sum_{i\neq j}^{N}\delta(|r_i - r_j| - r) \quad (2)$$

Here, r is the distance between two particles, V is the simulation box volume, $\delta(z)$ is the Dirac delta function, dr is the radial bin width, and $|r_i - r_j|$ is the distance between particles i and j. To compute the RDF, the simulation box is divided into concentric spherical shells of width dr. Particles within each shell are counted, and the result is normalized by the system's density to yield $g(r)$. The coordination number (CN), which indicates the number of particles in various solvation shells, is calculated from the RDF using the equation:

$$\mathrm{CN}(r) = 4\pi\rho N\int_0^r g(r)r^2 dr \quad (3)$$

In this equation, N is the total number of particles, ρ is the average particle density, which is referred to as the bulk density, and $g(r)$ is the RDF. The particle density is defined as the particle concentration at a specific distance r from a reference particle. The integration over the spherical shell centered on the reference particle provides a count of particles within a specified distance, thus determining the coordination number, which helps understand the local environment and interactions in the system.

2.2.3. Mean Squared Displacement. To evaluate self-diffusion, the slope derived from the MSD of molecules over time is plotted. The MSD represents the displacement of particles or molecules from their initial positions. It is calculated by squaring the difference between a particle's position at a given time and its initial position and then averaging this value across all particles in the system. The MSD is given by

$$\mathrm{MSD}(t) = \left\langle (r_i(t) - r_i(0))^2 \right\rangle = \frac{1}{N}\sum_{i=1}^{N}[(r_i(t) - r_i(0))^2] \quad (4)$$

Here, $r_i(t)$ is the position of atom i at time t, $r_i(0)$ is its initial position, and the angled brackets $\langle\rangle$ indicate an ensemble average over all particles. Using the Einstein relation, the diffusion coefficient D is determined from the MSD.

$$\beta = \frac{d[\log \mathrm{MSD}(t)]}{d[\log(t)]} \quad (5)$$

$$D = \frac{1}{6N}\lim_{t\to\infty}\frac{d}{dt}\sum_{i=1}^{N}[(r_i(t) - r_i(0))^2] \quad (6)$$

Here, β is the logarithmic slope of MSD. If the β value is less than 1, the behavior is subdiffusive; if the β value is greater than 1, then it is superdiffusive; and if the β value is 1, then it is diffusive.[40] This relationship allows the calculation of the diffusion coefficient from the MSD over time.

3. RESULTS AND DISCUSSION

3.1. Structural Orientation through RDF. Figure 1 represents the RDF analysis of the interaction between the oxygen moiety of caffeine and the potassium moiety of Acesulfame-K. The RDF for oxygen (denoted by $b_1\cdots n_1$ in Figure 1) indicates a first solvation shell at 7.25 Å with a coordination number *CN* of 4.26, suggesting that the oxygen atoms in caffeine establish more intense nonbonded interactions with the potassium atoms of Acesulfame-K compared to their nitrogen counterparts. The peak values of the RDF highlight that oxygen atoms play a significant role in the solubility and stability of the caffeine−Acesulfame-K mixture, primarily through long-range interactions, emphasizing the importance of oxygen in the molecular behavior of these compounds in solutions. Figure S1a−c of the Supporting Information depicts the RDF analysis for the interactions between various nitrogen moieties in caffeine and the potassium moiety of Acesulfame-K. Figure S2a shows the RDF for the nitrogen atom labeled as N2 in caffeine relative to the potassium of Acesulfame-K (denoted by $a_1\cdots n_1$), with the

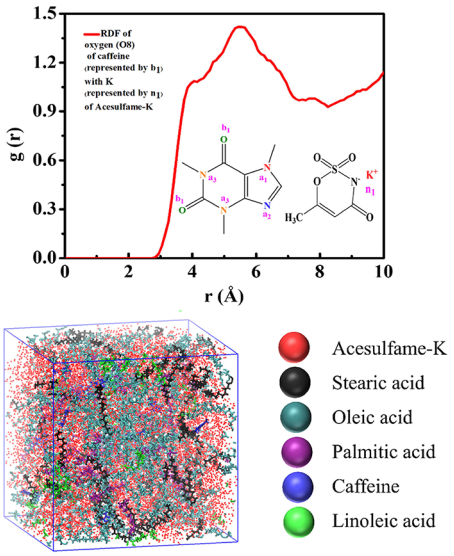


Figure 1. RDF of caffeine in Acesulfame-K, i.e., O8-K (denoted by $b_1\cdots n_1$). The snapshot of the simulation box is shown below.

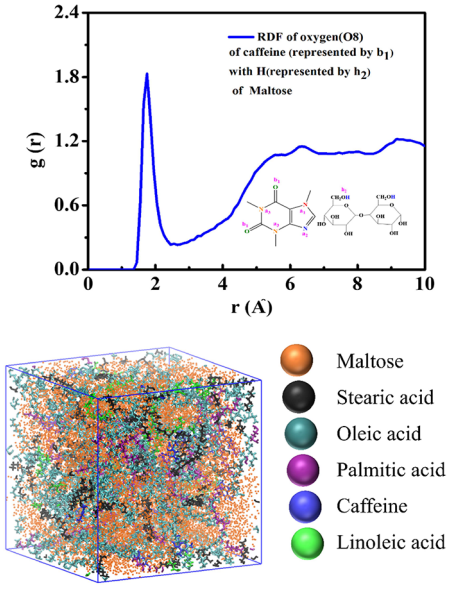


Figure 2. RDF of caffeine in Maltose, i.e., O8−H (denoted by $b_1\cdots h_2$). The snapshot of the simulation box is shown below.

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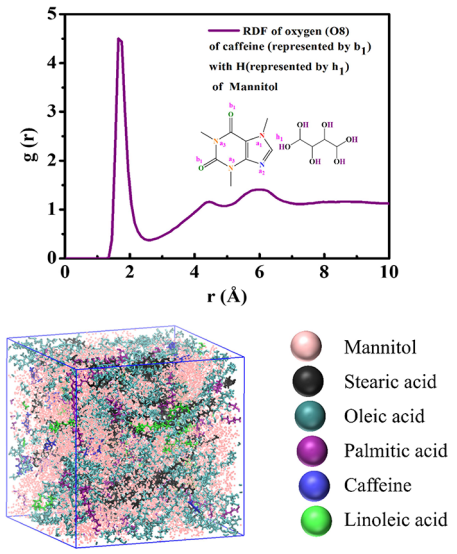


Figure 3. RDF of caffeine in Mannitol, i.e., O8−H (denoted by b_1···h_1). The snapshot of the simulation box is shown below.

first solvation shell observed at 4.05 Å. The presence of an immediate $-CH_3$ group adjacent to N2 causes the electron cloud to be less exposed, resulting in a lower interaction intensity.[41−43] Figure S2b presents the RDF for the nitrogen atom labeled as N4 in caffeine relative to the potassium of Acesulfame-K (denoted by a_2···n_1), with the first solvation shell appearing at 4.25 Å. The associated π-bonding with N4 (denoted by a2) reduces its electronegativity and lowers the interaction strength with Acesulfame-K. Figure S2c illustrates the RDF for the nitrogen atom labeled as N9 in caffeine relative to the potassium of Acesulfame-K (denoted by a_3···n_1), with the first solvation shell at 4.95 Å and a CN of 0.76. For all nitrogen-involved RDFs, the subsequent solvation shells are more visible than the first, suggesting that the interactions are primarily long-range. The differences in the RDF peaks and solvation shell distances highlight the varying interaction intensities and distances among the nitrogen moieties of caffeine with the potassium of Acesulfame-K, providing a detailed understanding of how the structural positioning and electronic environment of nitrogen atoms in caffeine affect their interaction with Acesulfame-K.

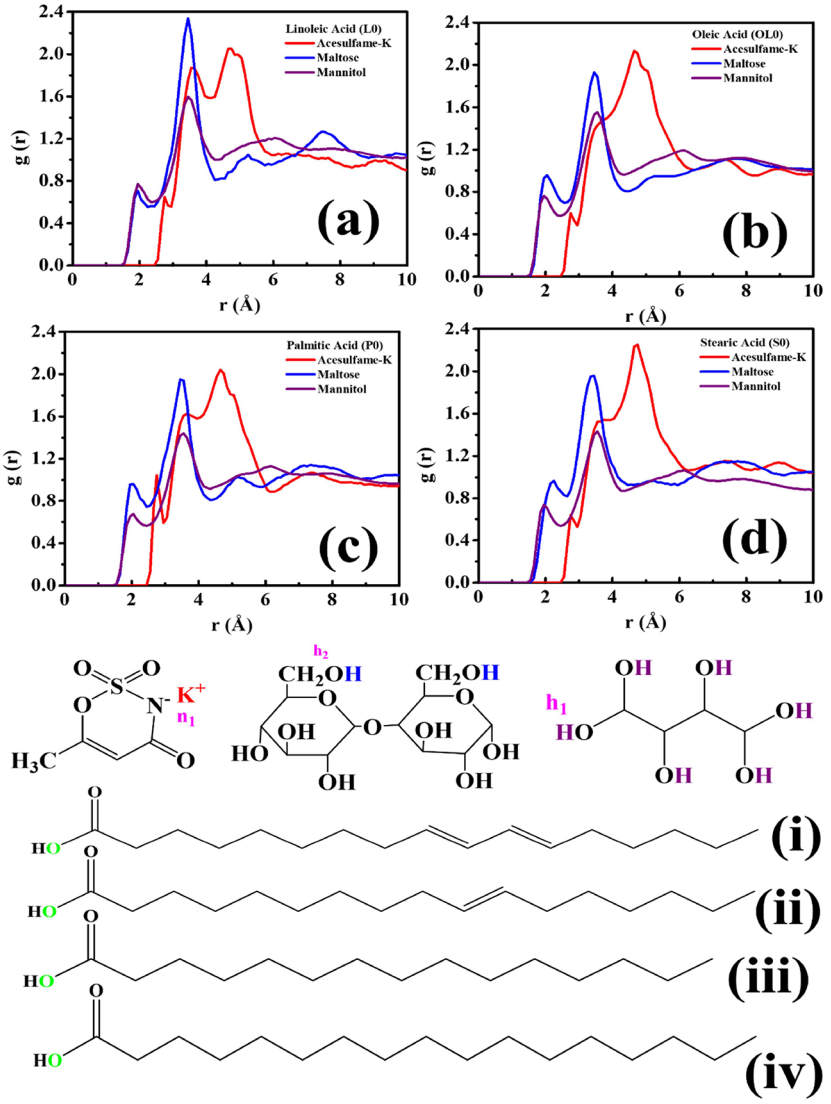


Figure 4. RDF of (a) linoleic acid, (b) oleic acid, (c) palmitic acid, and (d) stearic acid in Acesulfame-K (represented by red), Maltose (represented by blue), and Mannitol (represented by purple): (i) linoleic acid, (ii) oleic acid, (iii) palmitic acid, and (iv) stearic acid.

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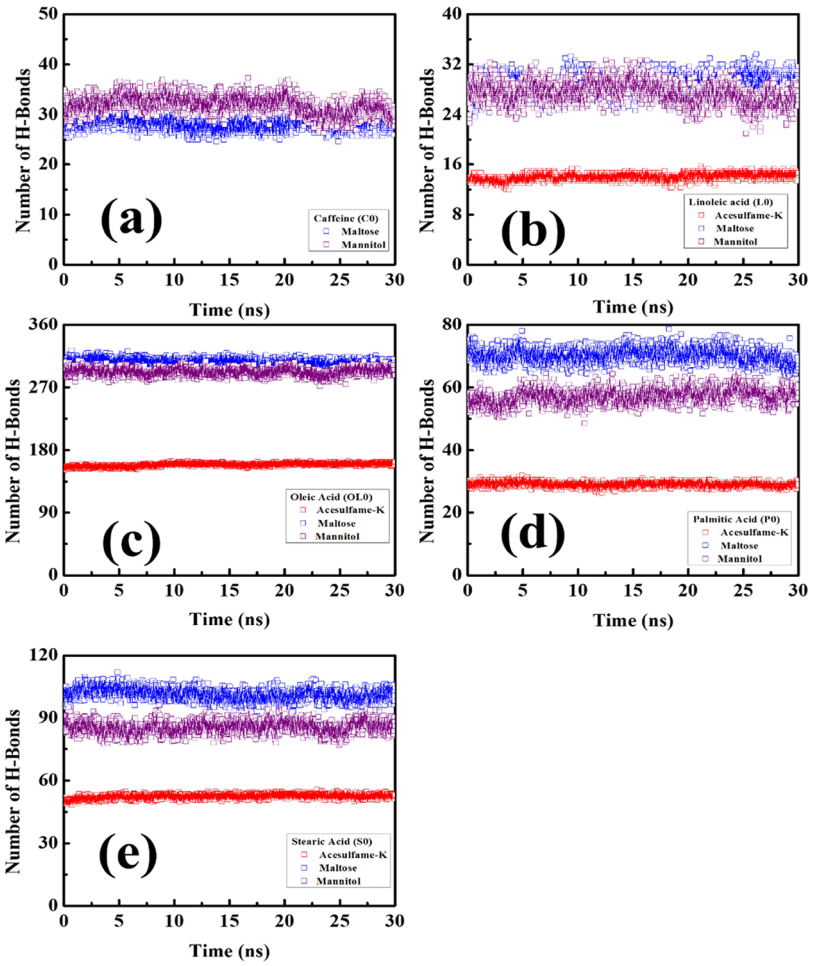


Figure 5. Hydrogen bonding of (a) caffeine, (b) linoleic acid, (c) oleic acid, (d) palmitic acid, and (e) stearic acid in Acesulfame-K (represented by red), Maltose (represented by blue), and Mannitol (represented by purple).

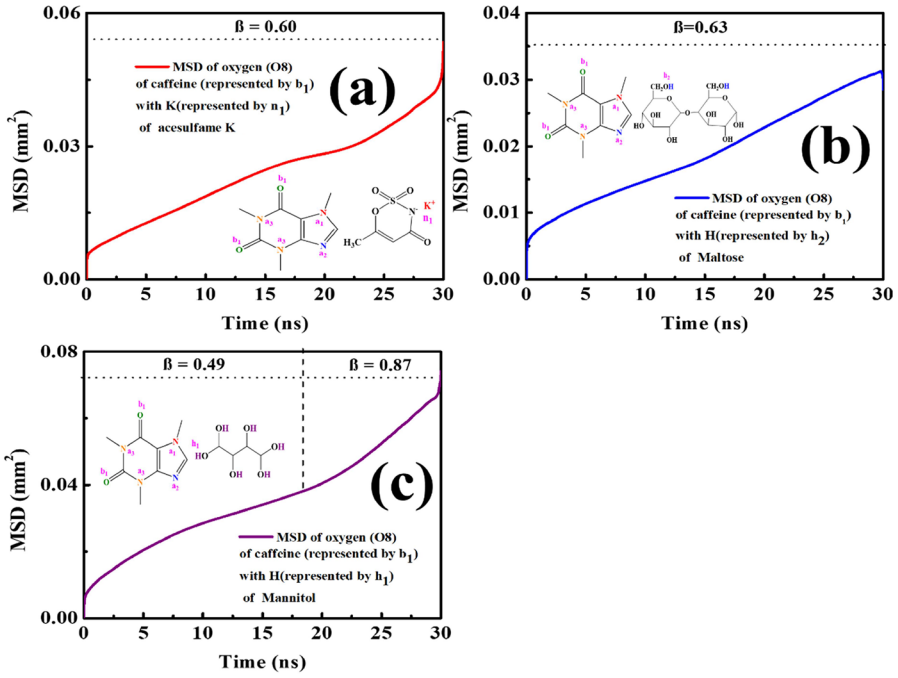


Figure 6. MSD of caffeine in (a) Acesulfame-K, (b) Maltose, and (c) Mannitol.

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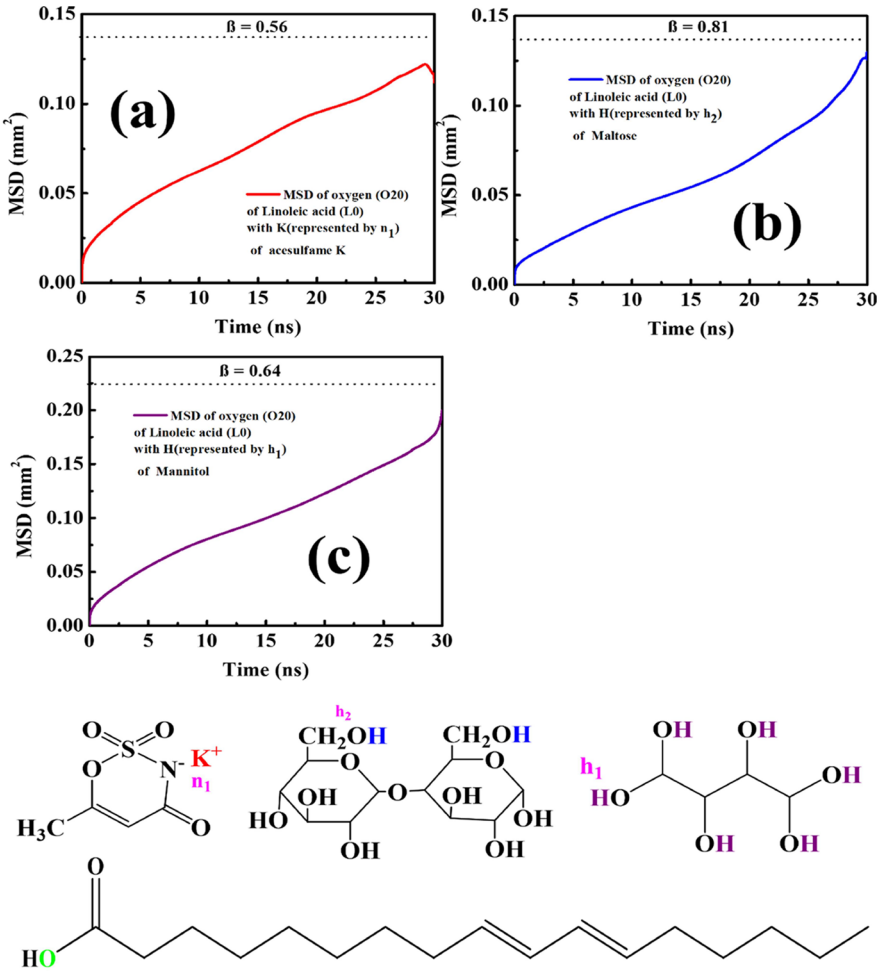


Figure 7. MSD of linoleic acid in (a) Acesulfame-K, (b) Maltose, and (c) Mannitol.

The RDF analysis for the oxygen atom represented by Figure 2 as (b_1) in caffeine shows a significant interaction with Maltose. The higher electronegativity of oxygen compared to nitrogen results in stronger hydrogen-bonding interactions, with a first solvation shell at 2.95 Å and a CN of 0.106. This is reflected in the pronounced RDF peak for oxygen, indicating that the electron density around the oxygen atom facilitates robust hydrogen bond formation with the hydroxyl moieties in Maltose. This interaction is crucial as it suggests that oxygen plays a major role in the molecular stability and solubility of the caffeine−Maltose complex. The strong interaction with oxygen enhances the overall binding affinity between caffeine and Maltose, leading to potentially more stable complexes in aqueous environments. The RDF profiles for the nitrogen atoms in Figure S3 of the Supporting Information (N2, N4, and N9 represented by a_1, a_2, and a_3, respectively) in caffeine reveal varying interaction strengths with Maltose. The RDF for N2 (a_1) is relatively lower due to the shielding effect of the adjacent $-CH_3$ moiety, which hinders the approach of Maltose molecules and weakens the hydrogen-bonding interactions. Conversely, N9 exhibits a more prominent peak because of its proximity to the oxygen atom. This proximity allows for shared hydrogen bonding, making the first solvation peak for a_3 more pronounced than for N2 or N4. For N4, the presence of a π-bond within the caffeine molecule shares the electron cloud toward the ring, making N4 less electronegative and further reducing its interaction with Maltose; the first solvation shell is observed at 3.15 Å with a CN of 0.16. The lower RDF peak for N4 indicates weaker hydrogen bonding compared to N9. These findings highlight the differential impact of structural shielding and electron distribution on the interaction strengths of various nitrogen atoms in caffeine with Maltose.

The RDF analysis in Figure 3 shows that the oxygen atom b_1 in caffeine has a significant interaction with Mannitol, with a first solvation shell at 2.75 Å and a CN of 0.106. This strong interaction, due to oxygen's higher electronegativity, enhances the molecular stability and solubility of the caffeine−Mannitol complex. The RDF profiles for the nitrogen atoms (N2, N4, and N9 represented in Figure S4 of the Supporting Information) reveal varying interaction strengths. N2 (denoted by a_1) shows a lower interaction due to the shielding effect of the $-CH_3$ moiety. N9 (denoted by a_3) exhibits a more pronounced peak due to its proximity to the oxygen atom (denoted by b_1), allowing for shared hydrogen bonding. N4 (denoted by a_2), influenced by a double bond, shows reduced interaction, with a first solvation shell at 3.05 Å and a CN of 0.74. These differences highlight the impact of structural shielding and electron distribution on nitrogen's interaction with Mannitol.

The RDF analysis in Figure 4 provides insights into the interactions between the oxygen of the −OH moiety of various fatty acids and the functional moieties of three sugars Acesulfame-K, Maltose, and Mannitol (represented by red, blue, and purple color, respectively). In Figure 4a, which

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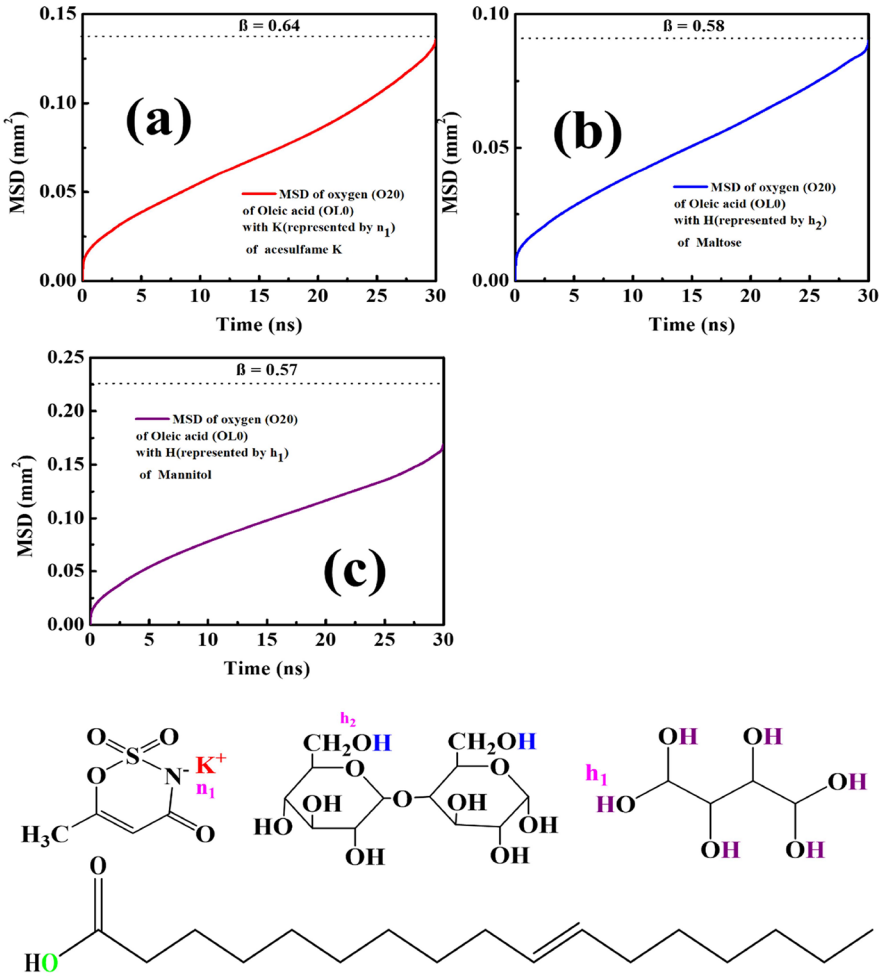


Figure 8. MSD of oleic acid in (a) Acesulfame-K (b) Maltose, and (c) Mannitol.

illustrates the interaction with linoleic acid (L0), the RDF shows that the first peak is lower than the second peak, indicating weak interactions between linoleic acid and the sugars. The coordination numbers CNs for the first solvation shell are 0.073 for Acesulfame-K, 0.099 for Maltose, and 0.341 Å for Mannitol, whereas the CNs for the second solvation shell are significantly higher at 0.72, 1.04, and 4.24, respectively, suggesting that electron cloud shielding hinders the nonbonded interactions. While the nonbonded interaction for Acesulfame-K is vdW in nature, and for the other two variants of sugar, the H-bond interaction prevails. The sharpest peak is observed for Maltose, indicating a comparatively intense interaction with linoleic acid. Figure 4b, showing the interaction with oleic acid (OL0), reveals that the first solvation shell appears at 3.05 Å for Acesulfame-K, 2.75 Å for Maltose, and 2.45 Å for Mannitol, with CN values of 0.065, 0.141, and 0.52, respectively, reflecting relatively weak interactions. The more pronounced second solvation shell underscores the significance of long-range interactions and the hindrance of short-range interactions such as vdW and H-bonds due to electron cloud shielding, with Acesulfame-K covering more area, suggesting a stronger interaction. Figure 4c details the interactions of the oxygen atoms in palmitic acid (PA0) with the potassium of Acesulfame-K, hydrogen of the $-CH_2OH$ moiety of Maltose, and hydrogen of the $-OH$ moiety of Mannitol. The RDF shows a sharp and short first peak accompanied by a pronounced second peak, with the first solvation shell observed at 3.05 Å for Acesulfame-K and 2.65 Å for both Maltose and Mannitol, with CNs of 0.094, 0.11, and 0.426, respectively. However, the second peak for Acesulfame-K, Maltose, and Mannitol is observed at 4.05, 4.25, and 4.45 Å with *CN*s of 0.68, 0.829, and 4.17, respectively. Figure 4d presents the RDF for interactions between the oxygen atoms in stearic acid (S0) and the sugars, showing a sharp and short first peak due to electron cloud shielding and a pronounced second peak, with the first solvation shell appearing at 3.05 Å for Acesulfame-K, 2.55 Å for Maltose, and 2.45 Å for Mannitol, with CNs of 0.069, 0.093, and 0.321, respectively. However, the second peak for Acesulfame-K, Maltose, and Mannitol is observed at 6.05, 4.25, and 4.25 Å with CNs of 3.21, 0.87, and 3.56, respectively. The RDF analysis indicates that the interactions between the oxygen of the $-OH$ moieties of fatty acids and the sugars are characterized by an initial weak interaction, followed by more significant long-range interactions. Acesulfame-K shows strong long-range interactions with fatty acids, as evidenced by the larger area covered and higher CNs in the second solvation shell. Maltose exhibits the sharpest peaks, indicating intense interactions with linoleic acid. Mannitol, while showing the least interaction in the first solvation shell, has substantial long-range interactions, particularly in linoleic acid. These insights are crucial for understanding the molecular interactions in chocolate bars, where these sugars and fatty acids are significant components,

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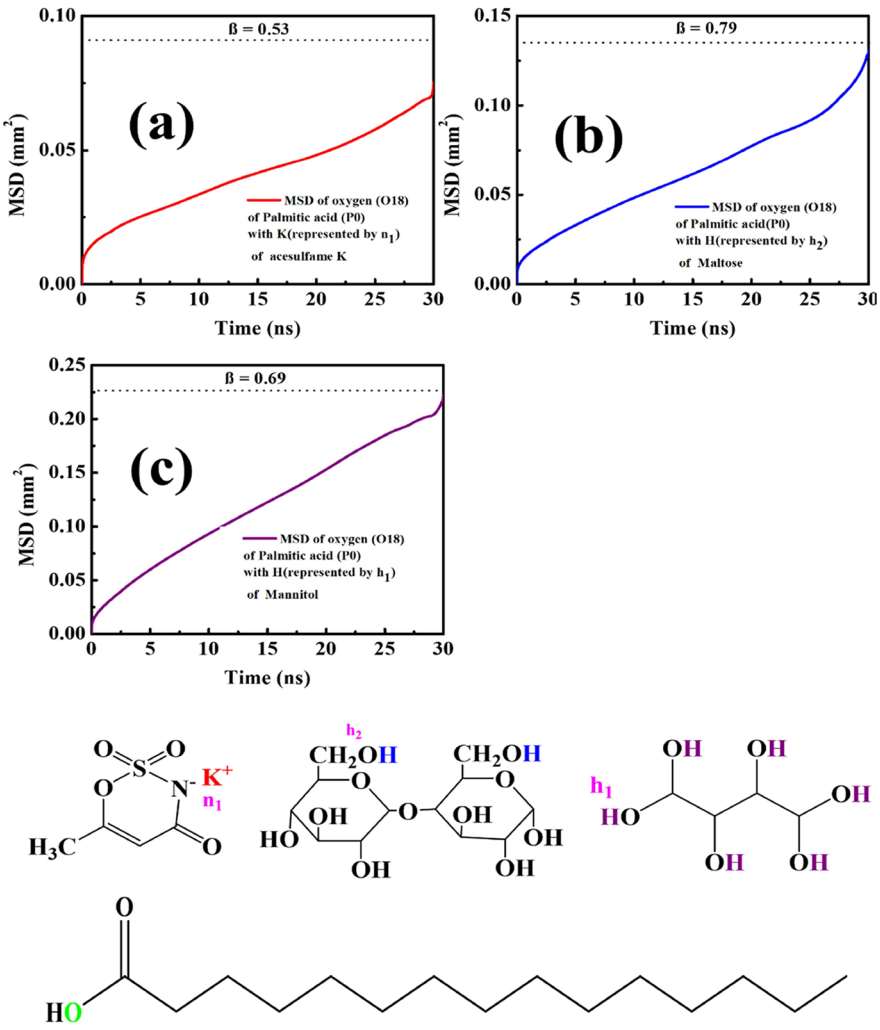


Figure 9. MSD of palmitic acid in (a) Acesulfame-K (b) Maltose, and (c) Mannitol.

influencing their physicochemical properties and potential applications in food and pharmaceutical formulations.

3.2. Evolution of Hydrogen-Bonding Network. The plot of H-bond formation of caffeine with various sugars is given in Figure 5a. Although there is no H-bond formation between caffeine and Acesulfame-K, caffeine exhibits stronger H-bonding with Mannitol (represented by purple), indicating a more intense and specific interaction. Maltose (represented by blue) also forms H-bonds with caffeine; however, the interaction strength is lower compared to Mannitol. These H-bonds play a crucial role in determining the solubility, stability, and overall behavior of caffeine when mixed with these sugars, reflecting the complex interplay of molecular forces in such systems. The H-bonding interactions between linoleic acid and the sugars Acesulfame-K, Maltose, and Mannitol (represented in Figure 5b) are dynamically captured through H-bonding graphs versus time. Linoleic acid consistently forms H-bonds with Acesulfame-K; however, these interactions are relatively transient and weak, as indicated by lower average H-bond counts over time. Conversely, the H-bonding between linoleic acid and Maltose is more stable and frequent, with the plot showing persistent and higher H-bond counts, reflecting stronger and more consistent interactions. Mannitol exhibits the strongest and most consistent H-bonding with linoleic acid, indicating robust and enduring H-bonds. These dynamic profiles underscore the varying strengths and stabilities of H-bonds formed between linoleic acid and the different sugars, with Maltose and Mannitol demonstrating more significant and enduring interactions compared to Acesulfame-K. The time-resolved analysis highlights the influence of the molecular structure on the formation and stability of H-bonds in these systems. The H-bonding interactions between oleic acid and the sugars Acesulfame-K, Maltose, and Mannitol (shown in Figure 5c) are analyzed by plotting H-bond counts over time. The oleic acid system exhibits a higher number of H-bonds compared to the other systems due to the higher concentration of oleic acid in the simulation box, which is suggested by Table S1 of the Supporting Information. This increased concentration provides more H-bond donors and acceptors, resulting in a greater number of H-bonds. Oleic acid forms relatively strong and frequent H-bonds with Acesulfame-K, as indicated by higher average H-bond counts. In contrast, its interactions with Maltose are more stable and consistent with the plots showing regular peaks and higher average counts, suggesting stronger and more persistent bonding. Mannitol exhibits the most robust H-bonding with oleic acid, with the highest and most stable counts in the graphs, reflecting durable and significant interactions. These profiles highlight the varying strengths and stabilities of H-bonds between oleic acid and the sugars, with

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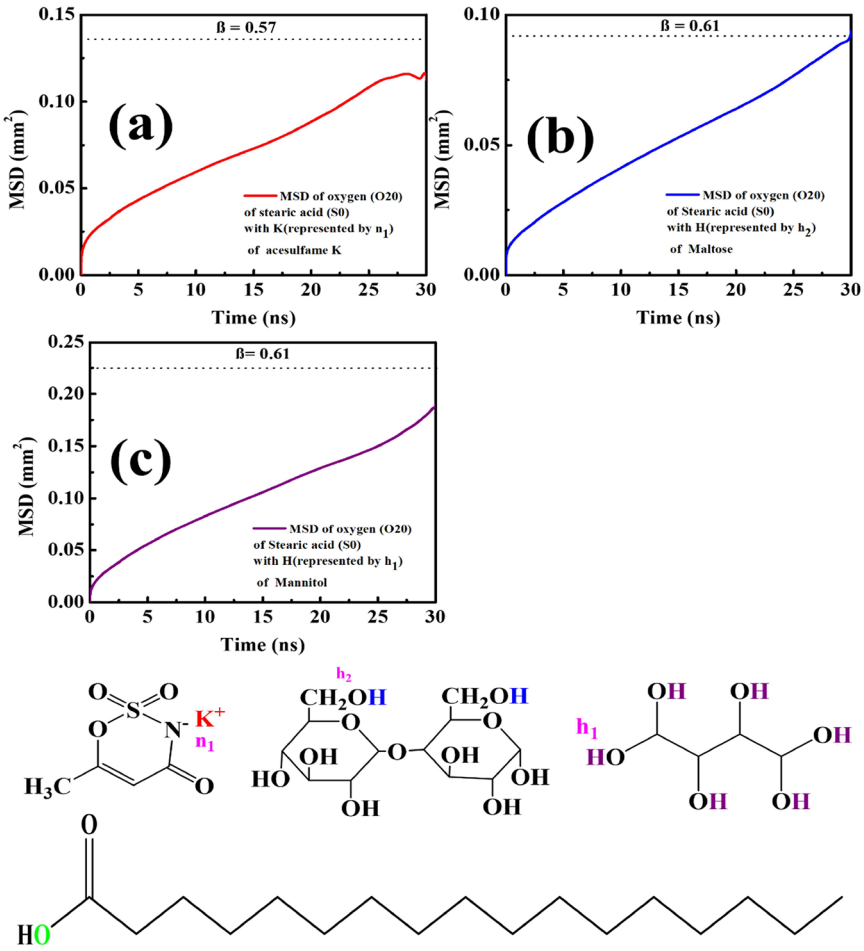


Figure 10. MSD of stearic acid in (a) Acesulfame-K, (b) Maltose, and (c) Mannitol.

Maltose and Mannitol demonstrating stronger and more enduring interactions compared to Acesulfame-K. The temporal evolution of H-bond formation for sugars when interacting with palmitic acid (PA0) is given in Figure 5d. The Acesulfame-K condition shows a relatively stable formation of H-bonds around an average of 30 throughout the 30 ns simulation period. The Maltose condition indicates a higher number of H-bonds, averaging around 70, demonstrating a stronger interaction strength and network formation. The Mannitol condition shows an average of around 60 H-bonds with minor fluctuations, suggesting a stable yet slightly less interactive molecular environment compared to the Maltose condition. The consistent number of H-bonds under all three conditions reflects the stability of the molecular structures. The Maltose condition exhibits a higher interaction capability, likely due to more favorable conditions for H-bond formation. These results highlight the varying stability and interaction strengths under different simulated conditions, providing insights into the MD and potential structural stability of the PA0 system with different sugars. The observed fluctuations are minimal, indicating a robust H-bond network formation that sustains over time, essential for understanding the behavior of these systems in practical applications. Conversely, stearic acid, another long-chain fatty acid, exhibits distinct H-bonding patterns with Acesulfame-K compared to palmitic acid in Figure 5e. This interaction is characterized by its hydrophobic tail and carboxyl moiety, creating stable bonds with Acesulfame-K molecules and influencing their dispersion in solutions. Stearic acid also interacts with Maltose and Mannitol through H-bonding, contributing to the physical properties of their mixtures, such as viscosity and crystallization behavior, pivotal in formulation sciences for various products. The results demonstrate the intricate molecular interactions among caffeine, fatty acids, and various sugars within the chocolate protein bar matrix. The H-bonding analysis reveals that caffeine forms stable hydrogen bonds with both saturated and unsaturated fatty acids present in the chocolate protein bar. These interactions are crucial for the stability and homogeneity of the chocolate protein bar, as H-bonds can significantly impact the physical properties of the product, such as its melting point and texture.[17] Additionally, the analysis shows that Acesulfame-K, due to its high-intensity sweetness and stability, forms stronger hydrogen bonds with caffeine compared to Maltose and Mannitol. This stronger interaction could potentially enhance the stability of caffeine within the chocolate matrix, preventing its rapid release and ensuring a sustained stimulant effect for consumers.[44]

3.3. Transport of Caffeine and Fatty Acids. The distributions of leading functional moieties for caffeine and fatty acids are established in the RDF. However, their time-resolved mobilities are visualized from the MSD of functional moieties. The diffusive behavior of caffeine and other fatty acids with Acesulfame-K, Maltose, and Mannitol visualized by red, blue, and purple plots, respectively, is evaluated by MSD. The results obtained from the simulations provide insights into the MSD of caffeine interacting with Acesulfame-K (repre-

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sented in Figures 6a and S5 of the Supporting Information), specifically analyzing the interaction at sites N2, N4, N9, and O8 (denoted by a_1, a_2, a_3, and b_1 respectively) of the caffeine molecule. The MSD plots, which measure the diffusion of molecules over time, are presented for each interaction site. The O8 site (denoted by b_1) displays the lowest β value of 0.60, reflecting the least molecular mobility and suggesting the strongest and most stable interaction with Acesulfame-K. The O8 site has the least movement, indicating the strongest and most stable interaction with Acesulfame-K. This means that different parts of the caffeine molecule bind differently to Acesulfame-K, with the O8 site being the most important for strong binding. The plot for the N2 (represented by Figure S5a and denoted by a_1) site shows a steady increase in MSD with a β value of 0.70, indicating a moderate level of molecular mobility and interaction stability over the 30 ns simulation period. The N4 (represented by Figure S5b and denoted by a_2) site exhibits a higher β value of 0.77, suggesting greater mobility and potentially less stable interactions compared with the N2 site. Similarly, the N9 (represented by Figure S5c and denoted by a_3) site shows a β value of 0.71, which aligns closely with the N2 site, indicating comparable interaction dynamics. The MSD plots show how caffeine molecules move when they interact with Acesulfame-K at different sites. For the N2 and N9 sites, the molecules move moderately, indicating stable interactions. The N4 site shows more movement, suggesting fewer stable interactions. Figures 6b and S5 of the Supporting Information illustrate the MSD of caffeine interacting with Maltose at specific sites N2, N4, N9, and O8 (denoted by a_1, a_2, a_3, and b_1, respectively). The O8 site, with a β value of 0.63, and the N2 site, with a β value of 0.55, show moderate molecular mobility, indicating a stable interaction. The N4 site, with a β value of 0.50, displays the least mobility, suggesting the most stable interaction. The N9 site has the highest β value of 0.67, indicating greater molecular movement and less stability. This reveals the subdiffusive behavior of all the moieties; also, the N4 site forms the strongest and most stable bond with Maltose, while the N9 site has the weakest interaction, providing valuable insights into the binding efficiency of caffeine with Maltose. Figure 6c presents the MSD analysis of caffeine interacting with Mannitol at site O8 over a 30 ns simulation period. The MSD plots indicate subdiffusive behavior across all sites, suggesting that caffeine−Mannitol interactions initially restrict molecular movement. Specifically, at the O8 site (Figure S6), the interaction starts with a β value, indicating strong initial binding, transitioning to more diffusive behavior over time, and reflecting a gradual loosening of the binding constraints. For the nitrogen sites, the analysis shows similar trends. At the N2 site (Figure S7a), the β value begins at 0.48, showing restricted movement, and later increases to 0.93, indicating a shift toward more diffusive behavior. At the N4 site (Figure S7c), the β value starts at 0.46 and transitions to 0.85. Finally, the N9 site (Figure S7b) also demonstrates a subdiffusive to diffusive transition, emphasizing strong initial binding, followed by gradual relaxation. This transition is observed due to the weakening of the structural constraints. These findings highlight the significance of site-specific interactions in determining the overall stability and mobility within the caffeine−Mannitol system.

Figure 7 represents the MSD analysis of the oxygen of the −OH moiety of linoleic acid interacting with Acesulfame-K, Maltose, and Mannitol over a 30 ns simulation period. The MSD plots consistently show subdiffusive behavior, indicating that the initial interactions restrict molecular movement. For instance, the interaction with Acesulfame-K exhibits an initial β value of 0.56, suggesting restricted movement, while the interaction with Maltose shows an initial β value of 0.81, indicating a stronger initial binding. The interaction with Mannitol has an initial β value of 0.64, reflecting a moderate restriction. As the simulation progresses, a transition toward more diffusive behavior is observed, reflecting a gradual loosening of the binding constraints. This subdiffusive nature suggests strong initial binding between the −OH moiety of linoleic acid and these molecules. This behavior is significant as it underscores the role of molecular mobility in determining the stability and strength of these interactions. Likewise, the subdiffusive regime of diffusion obtained with the −OH moiety of oleic acid, palmitic acid, and stearic acid interacting with Acesulfame-K, Maltose, and Mannitol is represented in Figures 8−10, respectively. As the simulation progresses, the MSD plots show a transition to more diffusive behavior, reflecting a loosening of these binding constraints over time. This transition from subdiffusive to diffusive motion is important from a technical perspective as it highlights the dynamic nature of molecular interactions. The subdiffusive behavior at the initial stage implies a stable complex formation, which gradually becomes more flexible as the interaction dynamics evolve. The varying β values across different interactions emphasize the differences in binding strength and stability, providing valuable insights into the molecular behavior of these systems.

4. CONCLUSIONS

This work uncovers the significant findings and implications of incorporating caffeine into chocolate protein bars, focusing on its interactions with fatty acids from cocoa powder and various sweeteners such as Acesulfame-K, Maltose, and Mannitol. Through MD simulations, detailed insights into the molecular behavior and interactions at the atomic level are gained, which are crucial for optimizing the formulation and stability of the final product. RDF analysis further supports these findings by providing a detailed spatial distribution of the molecules. The RDF plots indicate that caffeine molecules are more closely associated with the fatty acids in the presence of Acesulfame-K compared with Maltose and Mannitol. This close association suggests a more uniform distribution of caffeine throughout the chocolate matrix, which is essential for maintaining the quality and efficacy of chocolate protein bars. Moreover, the MSD analysis highlights the mobility differences in caffeine in the presence of different sugars. It is observed that the mobility of caffeine is most restricted with Acesulfame-K, indicating a more stable and less dynamic system, whereas Maltose and Mannitol show higher mobility, reflecting fewer stable interactions. The simulations demonstrate that Acesulfame-K, with its high-intensity sweetness and stability, forms moderately strong interactions with caffeine, as indicated by the β values observed in the MSD plots. Maltose, a disaccharide, shows the most stable interactions, particularly at the N4 site of caffeine, suggesting that it could enhance the structural integrity of the product. Mannitol, known for its noncariogenic properties and cooling effect, shows moderate restriction in molecular movement and hence implies the balanced interaction strength. Strong hydrogen bonds between caffeine, fatty acids, and sweeteners contribute to a more cohesive structure, enhancing the product's stability. The dynamic

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nature of these interactions, influenced by processing conditions such as temperature and mixing, is crucial for optimizing the formulation process. The findings emphasize the importance of understanding molecular interactions to enhance the product's stability. By leveraging MD simulations, such interactions can be estimated in silico, thereby reducing the dependency on extensive experimental trials and accelerating product development. This study bridges a critical knowledge gap and offers a foundation for future research and innovation in functional food products.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jced.4c00665.

Detailed information about the initial system, RDF plot, and MSD plots of caffeine in all sugars (PDF)

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A.S.T.: Writing—review and editing, writing—original draft, validation, methodology, investigation, formal analysis, and data curation. R.D.: Writing—review and editing, validation, methodology, investigation, and formal analysis. D.K.: Conceptualization, supervision, software, resources, project administration, and funding acquisition.

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